

Tutorial Sheet 4
II B. Tech. (Common)
Mathematics-II
(MCI102)

1. Consider the vector space P_2 consisting of all polynomials of degree at most two together with the zero polynomial. Let $S = \{-t^2 + 3, -2t^2 - 2t + 3\}$ be a set of polynomials in P_2 . Show that the set S is linearly independent in P_2 .
2. Compute the wronskian of the following sets of functions and state whether or not the two functions are linearly independent:
(a) $\{e^x, e^x \sin x\}$ (b) $\{x \ln |x|^2, x^2 \ln |x|^2\}, x \neq 0$ (c) $\{1, e^x\}$ (d) $\{e^x, \sin x, \cos x\}$ (e) $\{e^x, e^{5+x}\}$ (f) $\{x, x^2, x^3\}$
(g) $\{\ln x, \ln x^2, x\}, x > 0$ (h) $\{x, x^2, \sin x\}$ (i) $\{e^{mx}, xe^{mx}, x^2e^{mx}\}$.
Ans: (a) LI (b) LI (c) LI (d) LI (e) LD (f) LI (g) LD (h) LI (i) LI.
3. Show that the set of functions $e^{m_1x}, e^{m_2x}, \dots, e^{m_nx}$, where $m_1, m_2, \dots, m_n$ are distinct real numbers, is linearly independent.
4. Are the following statements true or false?
(a) If $y_1(x)$ and $y_2(x)$ are linearly independent on an interval I , then they are linearly independent on any interval containing I .
(b) If $y_1(x)$ and $y_2(x)$ are linearly dependent on an interval I , then they are linearly dependent on any subinterval of I .
5. Let $y_1(x)$ and $y_2(x)$ be two solutions of $y'' + P(x)y' + Q(x)y = 0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I . Are the following statements true or false?
(a) If $y_1(x)$ and $y_2(x)$ are linearly independent on an interval I , they are linearly independent on any interval J contained in I .
(b) If $y_1(x)$ and $y_2(x)$ are linearly dependent on an interval I , they are linearly dependent on any interval J contained in or containing I .
6. Let $y_1(x)$ and $y_2(x)$ be two solutions of $y'' + P(x)y' + Q(x)y = 0$, where $x \in I = (a, b)$, $P(x)$ and $Q(x)$ are continuous functions on I . Let $W(x)$ be the Wronskian of these two solutions. Show that $W'(x) = -P(x)W(x)$. If $W(x_0) = 0$ for some $x_0 \in I$, then prove that $W(x) = 0$ for each $x \in I$.
7. Find the general solution of the following second order homogeneous differential equations:
(a) $y'' + 3y' - 10y = 0$ **Ans:** $y(x) = Ae^{-5x} + Be^{2x}$
(b) $2y'' - 3y' = 0$ **Ans:** $y(x) = C_1 + C_2e^{\frac{3}{2}x}$
8. Find the curve $y(x)$ through the origin for which $y'' = y'$ and the tangent at the origin is $y = x$.
9. Find the differential equation of the form $y'' + ay' + by = 0$, where a and b are constants for which the following functions are solutions: (a) $e^{2x}, 1$ (b) $e^{-(\alpha+i\beta)x}, e^{-(\alpha-i\beta)x}$.
10. Find a homogeneous linear differential equation on $(0, \infty)$ whose general solution is $c_1x^2e^x + c_2x^3e^x$. Does there exist a homogeneous differential equation with constant coefficients with general solution $c_1x^2e^x + c_2x^3e^x$?
11. Solve the differential equation $4y'''' - 12y''' - y'' + 27y' - 18y = 0$.
Ans:
$$y(x) = Ae^x + Be^{2x} + Ce^{\frac{-3}{2}x} + De^{\frac{3}{2}x}$$
12. Solve the initial value problem $y''' + 3y'' - 4y = 0$, $y(0) = 1$, $y'(0) = 0$, $y''(0) = \frac{1}{2}$.
Ans:
$$y(x) = \frac{[e^x + (x+1)e^{-2x}]}{2}$$
13. Three solutions of a certain second order non-homogeneous linear differential equation are $y_1(x) = 1 + e^{x^2}$, $y_2(x) = 1 + xe^{x^2}$, $y_3(x) = (1+x)e^{x^2} - 1$. Find the general solution of the equation.
14. Solve $y'' - 4y = \cosh(2x - 1) + 3^x$.
Ans:
$$y = c_1e^{2x} + c_2e^{-2x} + \frac{x}{4} \sinh(2x - 1) + \frac{3^x}{(\log 3)^2 - 4}$$

15. Solve $y''' - y'' - 6y' = 1 + x^2$.

Ans:

$$y = c_1 + c_2 e^{3x} + c_3 e^{-2x} - \frac{1}{108}(6x^3 - 3x^2 + 25x)$$

16. Solve the differential equation $y'''' + 16y'' = x^2 + 5$.

Ans:

$$y = c_1 + c_2 x + c_3 \cos 4x + c_4 \sin 4x + \frac{x^2}{64} \left(\frac{x^2}{3} + \frac{39}{4} \right)$$

17. Solve $y'''' - y = e^x \cos x$.

Ans:

$$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5} e^x \cos x$$

18. Solve the differential equation $y''' + 2y'' + y' = \sin^2 x$.

Ans:

$$y = c_1 + (c_2 x + c_3) e^{-x} + \frac{x}{2} + \frac{3 \sin 2x + 4 \cos 2x}{100}$$

19. Solve the differential equation $y''' - 3y'' + 4y' - 2y = e^x$.

Ans:

$$y = c_1 e^x + e^x (c_2 \cos x + c_3 \sin x)$$

20. Find the general solution of the equation $y'' - 2y' - 3y = 6e^{-x} - 8e^x$.

Ans:

$$y(x) = A e^{-x} + B e^{3x} - \frac{3}{2} x e^{-x} + 2e^x$$

21. Find the general solution of the equation $y'' + 9y = \cos 3x$.

Ans:

$$y(x) = A \cos 3x + B \sin 3x + \frac{1}{6} x \sin 3x$$

22. Find the general solution of the equation $y'' + 4y' + 4y = 12e^{-2x}$.

Ans:

$$y(x) = (Ax + B) e^{-2x} + 6x^2 e^{-2x}$$

23. Find the general solution of the equation $y'' - 4y' + 13y = 12e^{2x} \sin 3x$.

Ans:

$$y(x) = e^{2x} [A \cos 3x + B \sin 3x - 2x \cos 3x]$$

24. Solve $y'' + 3y' + 2y = x e^x \sin x$.

Ans:

$$y = c_1 e^{-x} + c_2 e^{-2x} - e^{-2x} \left[\frac{x}{10} (\sin x + \cos x) + \frac{1}{50} (2 \sin x - 5 \cos x) \right]$$

25. Find a particular solution for the differential equation $y'' - 4y = e^x + \sin 2x$.

Ans: $y(x) = -\frac{1}{3} e^x - \frac{1}{8} \sin 2x$

26. Find the general solution of differential equation $y'' + y = 32x^3$.

Ans: $y(x) = A \cos 3x + B \sin 3x + 32x(x^2 - 6)$.

27. Solve the differential equation $y'' + a^2 y = \sec ax$.

Ans:

$$y(x) = c_1 \cos ax + c_2 \sin ax + \left(\frac{x}{a} \sin ax + \frac{\ln(\cos ax)}{a^2} \cos ax \right)$$

28. Use variation of parameters to find the general solution to the differential equation $y'' - 9y = \frac{9x}{e^x}$.

Ans:

$$y(x) = C_1 e^{3x} + C_2 e^{-3x} + \frac{9}{8} e^x \left(-x + \frac{1}{4} \right)$$

29. Use variation of parameters to find the general solution of the differential equation $y'' + 4y = \tan 2x$.

Ans:

$$y(x) = C_1 \cos 2x + C_2 \sin 2x - \frac{1}{4} \log(\sec 2x + \tan 2x)$$

30. It is given that $y_1 = x$ and $y_2 = \frac{1}{x}$ are two linearly independent solutions of the associated homogeneous equation of $x^2 y'' + xy' - y = x$, $x \neq 0$. Find a particular integral and the general solution of the equation.

Ans:

$$y(x) = C_1^* x + \frac{1}{x} C_2 + \frac{x}{2} \ln |x|, \text{ where } C_1^* = C_1 - \frac{1}{4}$$

31. Using variation of parameters find the general solution of the equation $y'' + 3y' + 2y = 2e^x$.

Ans:

$$y(x) = C_1 e^{-x} + C_2 e^{-2x} + \frac{1}{3} e^x$$

32. Find fundamental set of solutions $\{y_1(x), y_2(x)\}$ of the equation $x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + 5y = 0$ in the interval $x > 0$.

Ans:

$$y_1(x) = \frac{1}{x} \cos(2 \ln x), y_2(x) = \frac{1}{x} \sin(2 \ln x)$$

33. Find the general solution of the second order non-homogeneous differential equation $x^2 y'' - 5xy' + 8y = 8x^6$.

Ans:

$$y(x) = C_1 x^2 + C_2 x^4 + x^6$$

34. Find the general solution of the equation $x^2 y'' + 5xy' + 3y = \log x$, $x > 0$.

Ans:

$$y(x) = \frac{A}{x} + \frac{B}{x^3} + \frac{1}{3} \log x - \frac{4}{3}$$